

EduGenie – Google Gemini Powered Learning Assistant

Requirement Analysis Document

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Project Overview: EduGenie is an AI-powered learning assistant that uses Google Gemini to help students ask questions, understand concepts, generate study material, and practice learning in a simple and interactive way.

1. Introduction

Requirement analysis identifies what EduGenie should do, who will use it, and what technical and quality requirements are needed to build the system. The analysis provides a clear foundation for designing, developing, and testing the project.

2. Problem Statement

Students often need quick and understandable explanations when studying. Searching across multiple resources can take time, and general learning material may not match an individual student's level. EduGenie addresses this need by providing an AI-based learning assistant through a simple interface.

3. Objectives

- Provide students with quick answers to learning-related questions.
- Explain difficult concepts in a simple and structured manner.
- Generate useful notes and practice questions.
- Provide an interactive learning experience using generative AI.
- Create an easy-to-use interface suitable for students.

4. Scope of the System

In Scope: Student questions, AI-generated explanations, study notes, practice questions, interactive conversation, and a simple user interface.

Out of Scope: The system is not intended to replace teachers, guarantee academic correctness for every response, or provide professional advice outside the learning context.

5. Stakeholders

Stakeholder	Role / Requirement
Students	Primary users who ask questions and use learning features.
Project Team	Design, develop, test, and document the system.
Faculty / Guide	Reviews the project and provides academic feedback.

Stakeholder	Role / Requirement
AI Service	Google Gemini provides generative AI capabilities.

6. Functional Requirements

ID	Requirement	Description
FR-01	User Input	The system shall allow users to enter a question or topic.
FR-02	AI Response	The system shall generate a relevant response using Gemini.
FR-03	Topic Explanation	The system shall provide understandable explanations for topics.
FR-04	Study Notes	The system should support generation of concise study notes.
FR-05	Practice Questions	The system should generate questions for practice and revision.
FR-06	Conversation	The system should allow follow-up questions within the learning interaction.
FR-07	Display	The system shall display user inputs and AI responses clearly.
FR-08	Error Handling	The system should show a suitable message when a request cannot be processed.

7. Non-Functional Requirements

ID	Requirement	Description
NFR-01	Usability	The interface should be simple, clear, and easy for students to navigate.
NFR-02	Performance	Responses should be displayed within a reasonable time depending on the AI service.
NFR-03	Reliability	The system should handle normal user requests consistently.
NFR-04	Security	API keys and sensitive configuration details should not be exposed in the user interface.
NFR-05	Scalability	The design should allow additional learning features to be added later.
NFR-06	Maintainability	Code and project components should be organized for easier maintenance.
NFR-07	Compatibility	The application should work with commonly used modern browsers or its selected runtime.

8. Hardware Requirements

Component	Minimum Requirement
Computer / Laptop	A system capable of running the development environment.
Processor	Modern dual-core processor or better.
RAM	At least 4 GB recommended; 8 GB preferred for development.
Storage	Sufficient free space for project files, dependencies, and development tools.
Internet	Required for accessing cloud-based AI services such as Gemini.

9. Software Requirements

Software / Service	Purpose
Operating System	Windows, Linux, or another supported development environment.
Programming Language / Framework	Selected technology for implementing the EduGenie application.
Google Gemini	Generative AI capability for learning responses.
Code Editor / IDE	Writing, testing, and maintaining project code.
Web Browser	Running and testing the user interface, if web-based.
Internet Connection	Communication with required online services.

10. User Requirements

- Users should be able to enter questions easily.
- Users should receive readable and understandable responses.
- Users should be able to ask follow-up questions.
- Users should be able to use the system without advanced technical knowledge.
- The interface should provide clear feedback when an error occurs.

11. System Workflow

Open EduGenie → Enter Question/Topic → Send Request → Request Processed → Gemini Generates Response → Response Displayed → User Asks Follow-up / Starts New Topic

12. Assumptions and Constraints

- The AI service requires a working internet connection.
- Response quality depends partly on the input provided by the user and the AI service.
- API usage may be subject to service limits, availability, and configuration.
- The application should be used as a learning aid and important academic information should be verified.

13. Expected Outcome

The completed EduGenie system should provide students with a simple AI-assisted learning environment where they can ask questions, understand concepts, generate study resources, and practice topics. The project demonstrates how generative AI can be integrated into an educational application.

14. Future Requirements

- Voice input and voice-based responses.
- Tamil and other multilingual support.
- Personalized study plans and progress tracking.
- Image-based question support.
- Expanded quiz and examination preparation features.

15. Conclusion

The requirement analysis establishes the functional, non-functional, hardware, software, and user requirements for EduGenie. These requirements can be used as a reference during system design, development, testing, and project documentation.